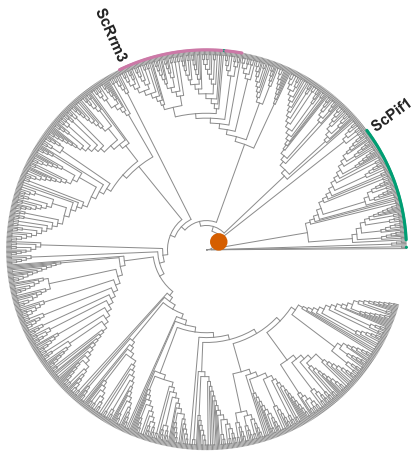


A

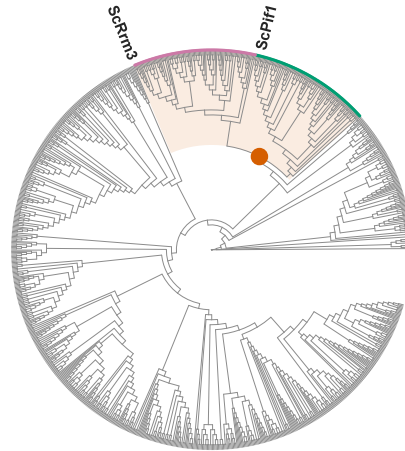
amino acids alone

MRCA -> 950 of 957 tips (21% Saccharomycotina)



amino acids + 3Di structure

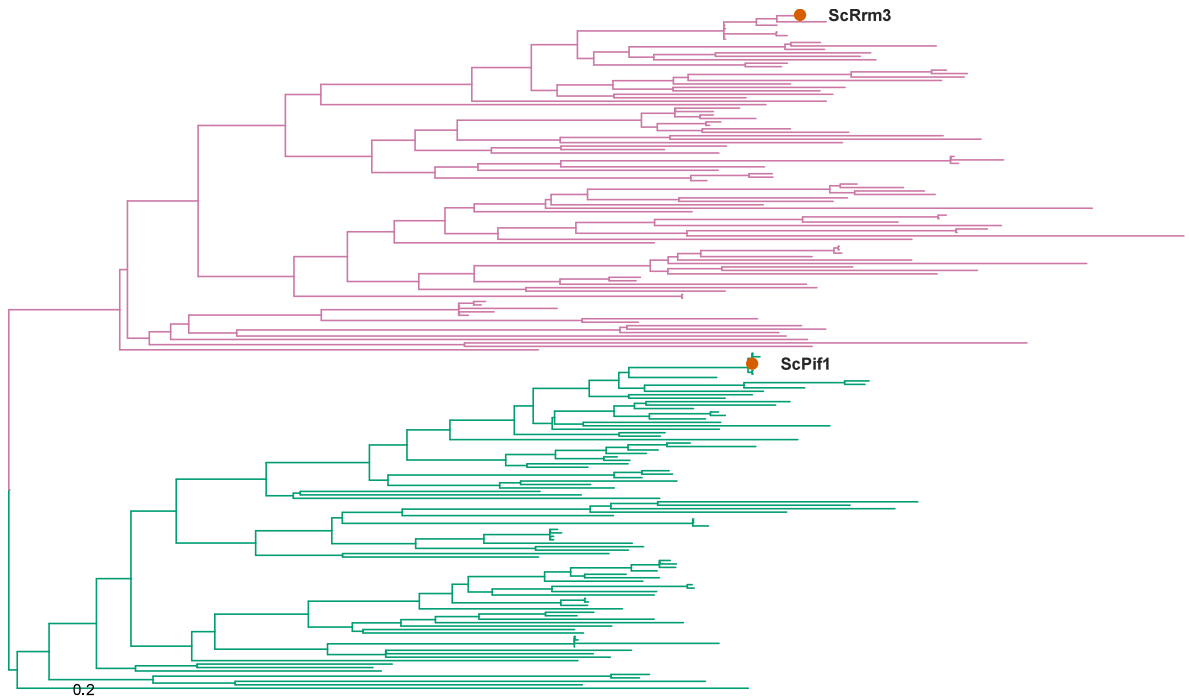
MRCA -> 197 of 957 tips (100% Saccharomycotina)



B

Branch lengths within the Saccharomycotina duplication clade

Rrm3 evolves 1.22x faster than Pif1 (root-to-tip 1.07 vs 0.88 subs/site) - the long-branch signal



● Pif1 clade ● Rrm3 clade ● Pif1/Rrm3 duplication

Human-rooted trees of 957 PIF1-family proteins; tips coloured by paralog (Pif1 vs Rrm3 clade).
3Di structure moves the duplication from Fungi to Saccharomycotina (A); Rrm3's long branches (B) are the LBA signal.